

Name of Student: Shreyasi Mishra	PRN: 25070122294
Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 3	
TITLE : Method Overloading and Static Members	

Problem Statement

Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Learning Objectives

To implement method overloading and use static variables and methods in Java programs.

Programme

1. Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.

```
public class Calculator {  
    static int calculationCount = 0;  
    int add(int a, int b) {  
        calculationCount++;  
        return a + b;  
    }  
    double add(double a, double b) {  
        calculationCount++;  
        return a + b;  
    }  
    public static void main(String[] args) {  
        Calculator calc = new Calculator();
```

```

        int intResult = calc.add(10, 20);

        System.out.println("Integer Addition: " + intResult);

        double doubleResult = calc.add(12.5, 7.8);

        System.out.println("Decimal Addition: " + doubleResult);

        System.out.println("Another Integer Addition: " + calc.add(50, 25));

        System.out.println("Total Calculations Performed: " + calculationCount);

    }
}

```

2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

```

public class RestaurantBilling {

    static int totalOrders = 0;

    double calculateBill(double amount) {

        totalOrders++;

        return amount + (amount * 0.05);

    }

    double calculateBill(double amount, boolean takeaway) {

        totalOrders++;

        return amount + (amount * 0.10);

    }

    double calculateBill(double amount, double deliveryCharge) {

        totalOrders++;

        return amount + deliveryCharge;

    }

    public static void main(String[] args) {

        RestaurantBilling rb = new RestaurantBilling();

        double dineIn = rb.calculateBill(500);
    }
}

```

```
        System.out.println("Dine-in Bill: Rs. " + dineIn);  
        double takeaway = rb.calculateBill(400, true);  
        System.out.println("Takeaway Bill: Rs. " + takeaway);  
        double delivery = rb.calculateBill(600, 50);  
        System.out.println("Delivery Bill: Rs. " + delivery);  
        System.out.println("Total Orders Processed: " + totalOrders);  
    }  
}
```

Output

Output

```
Integer Addition: 30  
Decimal Addition: 20.3  
Another Integer Addition: 75  
Total Calculations Performed: 3
```

Output

```
Dine-in Bill: Rs. 525.0  
Takeaway Bill: Rs. 440.0  
Delivery Bill: Rs. 650.0  
Total Orders Processed: 3
```

Conclusion

Through this assignment, we implemented method overloading and understood it as a form of compile-time polymorphism in Java, where multiple methods share the same name but differ in the number, type, or order of parameters, allowing the program to handle different inputs through a single method name. We also explored static variables and methods, demonstrating how they belong to the class itself rather than any individual object, meaning they are shared across all instances and can be accessed without creating an object. Overall, this assignment highlighted how method overloading and static members together contribute to cleaner, more flexible, and memory-efficient Java program design.